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| 2022 | <p>Q. 2. Discuss any two topics from the following:<br/> (a) The pinch effect in a plasma. (b) Electrical neutrality (Debye shielding) in a plasma.<br/> (c) Deduce the boundary conditions <math>B_{1n} = B_{2n}</math> and <math>E_{1t} = E_{2t}</math> from the Maxwell's equations.</p> <p>Q. 3. Discuss covariant form of electromagnetic equations.</p> <p>Q. 4. In a plasma, discuss the motion of charged particles in uniform electric and magnetic fields.</p> |
| 2021 | <p>Q.3. What is Ampere's law? Discuss the introduction of the displacement current. 10</p> <p>4. Find a transverse dispersion relation (from Maxwell's equations) for a homogeneous, isotropic and non-magnetic media with no prescribed charge and current distribution. 10</p> <p>Q.5. Discuss the reflection of a plane monochromatic wave from a conducting plane. 10</p>                                                                                            |
| 2020 | <p>Q. 3. What are plasma oscillations? Calculate plasma frequency using linearization process? (10)</p> <p>Q. 4. Describe the construction and working of PN junction laser? (10)</p> <p>Q. 5. Discuss the following topics (10)<br/> a) Wave equations for electric and magnetic fields. b) Magnetic mirror.</p>                                                                                                                                                        |
| 2019 | <p>Q.3 What are plasma oscillations? Calculate plasma frequency by using the linearization process. (10)</p> <p>Q.4 Discuss the construction and working of PN junction laser. (10)</p> <p>Q.5 Define skin depth. Discuss in detail the propagation of plane monochromatic waves in conducting media.(10)</p>                                                                                                                                                            |
| 2018 | <p>Q.3 Write brief note on following.<br/> (i) Power and efficiency of lasers (ii) Debye shielding in plasma. (10)</p> <p>Q.4 What is Ampere's law? Discuss the introduction of the displacement current. (10)</p> <p>Q.5 Discuss in detail the polarizations of EM waves. (10)</p>                                                                                                                                                                                      |
| 2017 | <p>Q1- Discuss in detail the polarization of electromagnetic waves. 10</p> <p>Q2- Define Debye Shielding ? What is Debye sphere? Calculate radius of Debye sphere. 10</p> <p>Q3- What is Ruby laser? Explain its construction and working in detail 10</p>                                                                                                                                                                                                               |
| 2016 | <p>Q1. What are plasma oscillations ? By help of linearization process ,calculate plasma frequency ? 10</p> <p>Q2. Discuss in detail reflection and refraction at the boundary of two non-conducting media ? 10</p> <p>Q3. Write a note on working Mechanism of Ruby Laser ? 10</p>                                                                                                                                                                                      |
| 2015 | <p>Q.1 Discuss the motion of a charge particle in plasma under the influence of constant electric and magnetic field . (10)</p> <p>Q.2 Discuss in the propagation of plane monochromatic waves in the conducting media (10)</p> <p>Q.3 Describe the construction and working of Ruby Laser . (10)</p>                                                                                                                                                                    |
| 2014 | <p>Q.1 Define skin depth .Discuss the propagation of plane monochromatic waves in conducting medium . (10)</p> <p>Q.2 Derive the expression of scalar and vector potential for the case of radiation from an oscillating dipole (10)</p> <p>Q.3 Discuss the construction and working of ruby laser . (10)</p>                                                                                                                                                            |

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| 2021 | <p>I. Define, unpolarized, plane (or linearly) polarized, vertically and horizontally polarized waves.</p> <p>II. Briefly discuss the concept of skin depth in the conducting media.</p> <p>III. To confine a plasma, discuss very briefly the concept of magnetic mirror.</p> <p>IV. What is circular and elliptical polarization?</p> <p>V. Discuss few applications of lasers.</p> <p>VI. What is difference between index of refraction and dielectric constant?</p>                                                                                  |
| 2021 | <p>1. What happens, when a constant a plasma electric field is applied to plasma? 4</p> <p>2. What is Pinch effect in plasma ? 3</p> <p>3. What do you mean by plasma oscillations ? 3</p> <p>4. What is difference between a neutral gas and a plasma? 4</p> <p>5. What is Poynting theorem? 4</p> <p>6. What is phase velocity of a wave? 2</p>                                                                                                                                                                                                         |
| 2020 | <p>a) Differentiate between neutral gas and plasma. (4)</p> <p>b) Discuss briefly the applications of laser. (4)</p> <p>c) What is difference between elliptical polarization and circular polarization? (5)</p> <p>d) Obtain covariant form of Lorentz condition. (4)</p> <p>e) Write down the criteria for an ionized gas to be plasma. (3)</p>                                                                                                                                                                                                         |
| 2019 | <p>i. Differentiate between a neutral gas and a plasma. 4</p> <p>ii. Discuss briefly the circular and elliptical polarizations. 5</p> <p>iii. What are the three criteria for an ionized gas to be a plasma? 3</p> <p>iv. Discuss briefly the few applications of laser? 4</p> <p>V. Calculate the frequency of sea water if skin depth is <math>\delta = 1</math>, <math>\mu = \mu_0</math> and Conductivity = <math>4.3 \text{ S/m}</math>. (4)</p>                                                                                                     |
| 2018 | <p>a) What is Pinch effect? 4</p> <p>b) Discuss few applications of lasers very briefly. 4</p> <p>c) Describe briefly the black body radiation? 3</p> <p>d) What is skin depth? 2</p> <p>e) What are retarded vector and scalar potentials? 4</p> <p>f) What is Poynting vector? 3</p>                                                                                                                                                                                                                                                                    |
| 2017 | <p>a) Describe briefly the Pinch effect. 5</p> <p>b) What is meant by a plane wave? 4</p> <p>c) What are right and left circularly polarized waves? 4</p> <p>d) What are black body radiations? 3</p> <p>e) What are retarded vector and scalar potentials. 4</p>                                                                                                                                                                                                                                                                                         |
| 2016 | <p>1. Derive covariant form of equation of continuity ? 3</p> <p>2. Write down criteria for plasma ? 3</p> <p>3. What is Circular polarization ? 3</p> <p>4. State Poynting theorem And also explain its terms . 4</p> <p>5. Write some applications of plasma ? 3</p> <p>6. Write down Maxwell's equation in differential and integral form ? Also explain physically?</p>                                                                                                                                                                               |
| 2015 | <p>1) What is plasma oscillation ? (3)</p> <p>2) What do you know between Giant pulse techniques ? (4)</p> <p>3) For a typically air glass interface, find reflectance and transmittance (<math>n = 1</math> for air and <math>n = 1.5</math> for glass ).</p> <p>4) What is retarded scalar potential ? (3)</p> <p>5) Calculate frequency of sea water if skin depth <math>\delta = 1</math>, <math>\mu = \mu_0</math> and conductivity <math>g = 4.3 \text{ S/m}</math> (3)</p> <p>6) Write down the criteria for an ionized gas to be plasma ? (3)</p> |
| 2014 | <p>a) Derive transverse dispersion relation for non-magnetic, non-conducting media . (3)</p> <p>b) Discuss briefly the giant pulse techniques. (3)</p> <p>c) What is elliptical polarization. (3)</p> <p>d) What is retarded scalar potential ? (3)</p> <p>e) What is difference between neutral gas and plasma ? (2)</p> <p>f) Mention any three applications of plasma ? (3)</p> <p>g) Write criteria for an ionized gas to be plasma . (3)</p>                                                                                                         |